

Extended Isolation Forest - Anomaly Detection Course

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Abstract

Various methods have been proposed to detect anomalies in datasets, one of them being the Isolation Forest which relies on the principle that anomalies require fewer splits to be isolated than normal data points. In this article, we will explore an enhanced version of the Isolation Forest algorithm, called Extended Isolation Forest, which keeps the simplicity of the original algorithm, but also improves its performance. We will also explore the performance of these algorithms on different datasets and measure their efficiency in detecting anomalies.

1 Introduction

Anomalies are data points that have different characteristics from the majority of the data and detecting anomalies is an important task in many fields, such as fraud detection, network security, astronomy, and many others. Among the various methods proposed to detect anomalies, we will focus on the Isolation Forest [2] algorithm and an enhanced version of it, Extended Isolation Forest [1].

Exploiting the fact that anomalies are "few and different", both Isolation Forest and Extended Isolation Forest algorithms are tracking the average path length a data point travels in a binary tree until it is isolated.

The main difference between the two algorithms consists in the way they are splitting the data points. While the Isolation Forest algorithm is choosing a random feature and a random value within the feature's range, basically choosing a hyperplane parallel to the axis, the Extended Isolation Forest algorithm generalizes this idea by choosing a random hyperplane, which can be at any angle.

2 Technical description of the problem

To better understand the problem, we will describe the splitting process of the Isolation Forest algorithm and the formula for calculating the anomaly score.

2.1 Standard Isolation Tree Splitting

The core idea behind the Isolation Tree is the recursive splitting of the data points in a binary tree as follows:

1. **Random Feature Selection:** Choose one feature at random from the dataset.
2. **Random Split Value:** Choose randomly a splitting value within the minimum and maximum range of the chosen feature.
3. **Partition:** Split the data into two sub trees by comparing the data points with the chosen split value.
4. **Recursion:** Repeat the process recursively until either the maximum tree depth is reached or there is only one data point in a node.

Formally, at any given node t , let q be the randomly chosen feature and p be the randomly chosen split value. A data point x is sent to the left child if $x_q < p$ and to the right child otherwise. Notice that the hyperplane formed by the split is parallel to the axis.

2.2 Anomaly Score Calculation

Once an ensemble of t Isolation Trees is formed, each tree i will return the path length $h_i(x)$ of a

data point x which represents the number of edges traversed by x until it is isolated. These values can be plugged into the following formula to calculate the anomaly score of a data point x :

$$s(x, n) = 2^{-\frac{E(h(x))}{c(n)}} \quad (1)$$

where $E(h(x))$ is the average path length of x in the ensemble of trees and $c(n)$ is the average path length of an unsuccessful search in a binary tree with n data points.

Intuitively, a lower value for $E(h(x))$ indicates that x was isolated faster, hence a higher value for $s(x, n)$, which means that x is more likely to be an anomaly.

2.3 Isolation Forest Limitations

Due to the fact that Isolation Tree splitting strategy is limited to hyperplanes parallel to the axis, the algorithm is forced to draw straight lines to separate the data points. This can lead to blurred boundaries between normal and anomalous data points and also add false positives to the anomaly detection.

Below are a few examples that illustrate the limitations of the Isolation Forest algorithm:

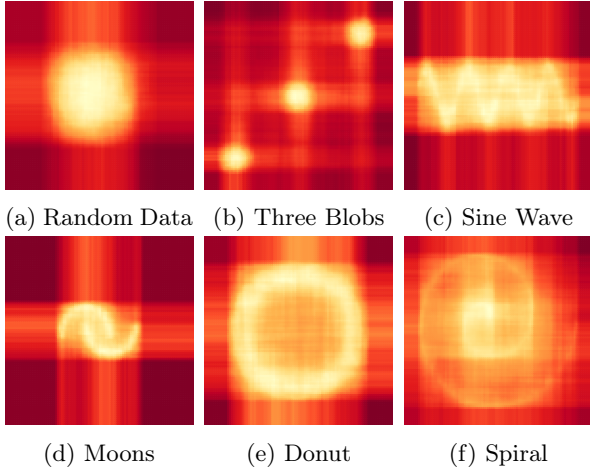


Figure 1: Isolation Forest on various data distributions

3 Possible solution

3.1 Extended Isolation Forest

To overcome the limitations of the Isolation Forest, we will explore the ideas behind the Extended Isolation Forest algorithm. In order to avoid the constraints of the axis-parallel hyperplanes, the Extended Isolation Forest splitting strategy will choose a random hyperplane at random angles.

Formally, at any given node t and knowing that there are d features, let $\vec{n}$ be a normal vector uniformly sampled from the unit sphere in R^d and p be an intercept uniformly sampled from the range of values present at each branching point. These values will determine the hyperplane that will split the data points into two sub trees.

Now choosing which side of the hyperplane a data point x will go is done by checking the sign of the following equation:

$$(\vec{x} - \vec{p}) \cdot \vec{n} < 0 \quad (2)$$

where $\vec{x}$ is the data point, $\vec{p}$ is the intercept, and $\vec{n}$ is the normal vector of the hyperplane.

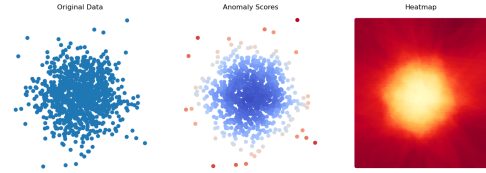


Figure 2: Extended Isolation Tree

3.2 Extension Level

Intuitively, the Extended Isolation Forest is a generalization of the Isolation Forest algorithm, in the sense that it can draw hyperplanes at any angle, not just parallel to the axis. We can call the number of axis parallel to the hyperplanes the "extension level" of the algorithm. Hence, in our 2D case, the Isolation Forest algorithm has an extension level of 1, while the Extended Isolation Forest algorithm has an extension level of 2. Actually, we can adjust the extension level to any value we want by zeroing out the normal vector components that we do not want to use.

Formally, we can zero out random features from the normal vector $\mathbf{n}$ according to the extension level e as follows:

$$\mathbf{n}_e = \mathbf{n} \odot \mathbf{m} \quad (3)$$

where $\mathbf{m}$ is a binary mask with d elements, where e elements are set to 1 and $d - e$ elements are set to 0.

4 Experiments and Results

4.1 Heatmaps

We have conducted experiments using artificially generated datasets with different distributions, such as random data, three clusters, sine wave, moons, donut, and spiral. We used $sample_size = 500$ and $n_trees = 100$ for the heatmaps in the figure 3

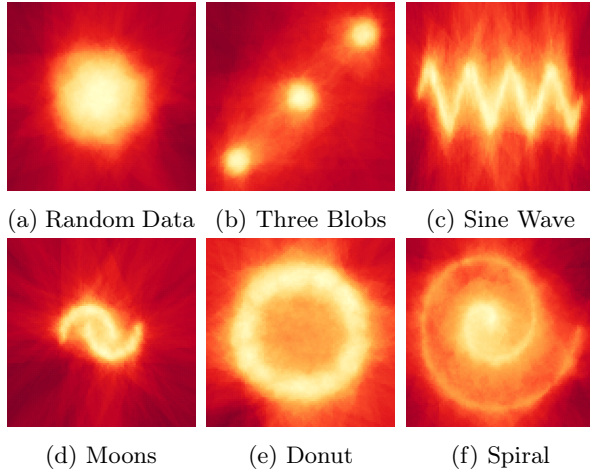


Figure 3: Extended Isolation Forest with 100 trees

Notice how the vertical and horizontal lines got smoothed out using the Extended Isolation Forest algorithm, which is a direct result of the hyperplanes being drawn at random angles. Looking closely at the heatmaps, we can still distinguish some fine lines due to the fact that we only used 100 trees in the ensemble. Increasing the number of trees $n_trees = 300$ will further smooth out the lines as shown in the figure 4.

Visually, we can see that those fine lines are gradually disappearing as we increase the number of trees in the ensemble.

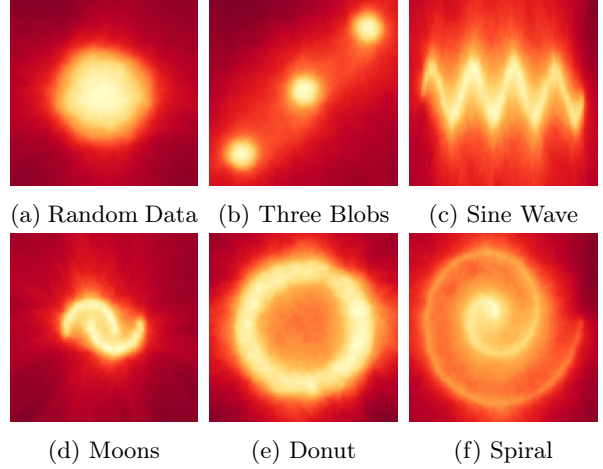


Figure 4: Extended Isolation Forest with 300 trees

4.2 Mean and Variance of Anomaly Scores

In order to quantify the performance of the Extended Isolation Forest, we took the one cluster dataset and drew concentric circles with different radii around the cluster and then calculated the mean and variance of the anomaly scores for each circle. This way, we can see how the anomaly scores are distributed around the cluster and quantify the performance of the algorithm. 5

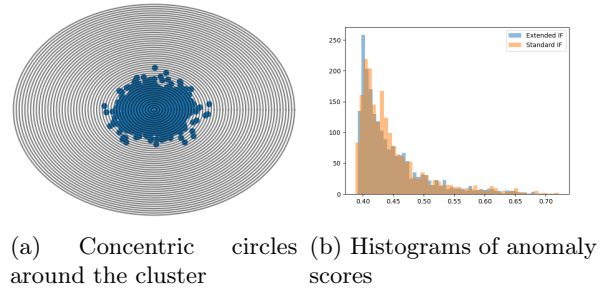
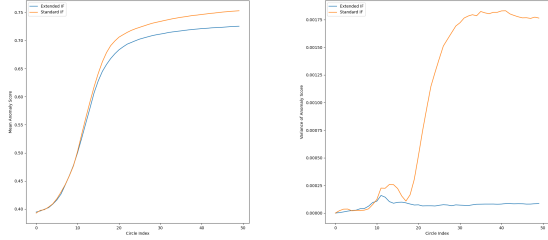


Figure 5: Concentric circles and data distribution

We drew 50 concentric circles around the generated cluster and for a fixed circle, we calculated the mean and variance of the anomaly scores for the data points that lie on the circle. The results are shown in the figure 6.

Notice how the mean anomaly scores increase abruptly as we move away from the center of the cluster and how the rate of increase slows down af-



(a) Mean Anomaly Scores (b) Var of Anomaly Scores

Figure 6: Mean and Variance of Anomaly Scores

terwards. Also, notice a significant improvement in the variance of the anomaly scores from the Extended Isolation Forest around the 20th circle.

4.3 AUC ROC / AUC PRC

Another metric worth mentioning is the Area Under the Receiver Operating Characteristic (AUC ROC) curve and the Area Under the Precision-Recall Curve (AUC PRC). Besides the data artificially generated we also added anomalies to the dataset and computed these metrics for both the Isolation Forest and the Extended Isolation Forest in the table 1.

Table 1: Performance comparison of Isolation Forest (iForest) and Extended Isolation Forest (EIF) on different datasets.

Data	AUC ROC		AUC PRC	
	iForest	EIF	iForest	EIF
Single Blob	0.932	0.935	0.810	0.834
Double Blob	0.971	0.974	0.892	0.930
Sinusoid	0.915	0.943	0.781	0.796
Moons	0.905	0.920	0.821	0.856
Donut	0.807	0.859	0.567	0.654
Spiral	0.780	0.806	0.415	0.463

Comparing the AUC ROC and AUC PRC values for the Isolation Forest and the Extended Isolation Forest, we can see that the Extended Isolation Forest consistently outperforms the Isolation Forest algorithm on all datasets. The improvement is more significant on datasets with complex distributions, such as the donut and spiral datasets.

5 Conclusion

In this article, we explored two algorithms for anomaly detection, the Isolation Forest and the Extended Isolation Forest. We have shown that with minimal adjustments to the splitting strategy of the Isolation Forest algorithm, we can significantly improve the performance of the algorithm. The generalization of the hyperplanes to any angle allows the Extended Isolation Forest to better capture the data distribution and avoid the limitations of the Isolation Forest algorithm.

Using the "extension level" parameter, we can control the complexity of the hyperplanes, even particularizing the Extended Isolation Forest to the Isolation Forest by setting the extension level to 1, which basically forces the hyperplanes to be parallel to the axis.

The results of the experiments support the idea that the Extended Isolation Forest algorithm is more robust and outperforms the Isolation Forest algorithm on various datasets.

References

- [1] Sahand Hariri, Matias Carrasco Kind, and Robert J. Brunner. Extended isolation forest. volume 33, page 1479–1489, April 2021.
- [2] Fei Tony Liu, Kai Ting, and Zhi-Hua Zhou. Isolation forest. pages 413 – 422, 01 2009.